


$$\text{BLP} \quad E^*(Y|X) = \alpha + \beta X$$

$$\alpha = E(Y) - \beta E(X)$$

$$\beta = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}$$

You need $f(X, Y)$

↑
In general you don't have this in "real life"

What is feasible?

↳ Observations on the joint vector (X, Y)

$(X_1, Y_1), (X_2, Y_2), \dots, (X_n, Y_n)$

↓ estimate the BLP coefficients

Least-Squares estimator $(\hat{\beta}_0, \hat{\beta}_1) = \underset{c_0, c_1}{\operatorname{argmin}} \frac{1}{n} \sum_{i=1}^n (Y_i - c_0 - c_1 X_i)^2$

↓ minimizing the sum of squared deviations

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$$\text{where } \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$$

$$f(x_1, y_1, x_2, y_2, \dots, x_n, y_n) = \prod_{i=1}^n f(x_i, y_i)$$

$$= f(x_1, y_1) \times f(x_2, y_2) \times \dots \times f(x_n, y_n)$$

Assumption 4

$$\text{Var}(Y_i | X_i) = \sigma^2 \quad \text{for all } i$$

Since I'm computing $\hat{\beta}_0, \hat{\beta}_1$ are also random variables

$$E(\hat{\beta}_1 | X's)$$

$$E(\hat{\beta}_0 | X's)$$

$$\text{Var}(\hat{\beta}_0, \hat{\beta}_1 | X's) = \frac{\sigma^2}{S_{XX}} \begin{bmatrix} \frac{1}{n} \sum_{i=1}^n X_i^2 & -\bar{X} \\ -\bar{X} & 1 \end{bmatrix}$$

$$\text{where } S_{XX} = \sum_{i=1}^n (X_i - \bar{X})^2$$

Suppose you have estimated $\hat{\beta}_0, \hat{\beta}_1$
The "errors" are

$$\hat{\epsilon}_i = Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i$$

An estimator of σ^2 is $\hat{\sigma}^2 = \frac{1}{n-2} \sum_{i=1}^n \hat{\epsilon}_i^2$

$$E(\hat{\sigma}^2 | X\text{'s}) = \sigma^2$$

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this makes $\hat{\sigma}^2$ unbiased.

In this context, linear means that

Regression Line $\hat{\beta}_0 + \hat{\beta}_1 X_i$ is linear in $\hat{\beta}_0$ and $\hat{\beta}_1$

Recall from lecture 2

BLP: $E^*(Y|X, X^2) = \beta_0 + \beta_1 X + \beta_2 X^2$ This is still linear in $(\beta_0, \beta_1, \beta_2)$

↓ Less trivial example of
a non-linear predictor

$$Y = \begin{cases} 1 \\ 0 \end{cases}$$

This is called a Probit Model and is non-linear

$$\begin{aligned} E[Y|X] &= P_r(Y=1|X) = \text{Normal}(\beta_0 + \beta_1 X, \sigma^2) \\ &= \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(y - \beta_0 - \beta_1 X)^2}{\sigma^2}\right) \end{aligned}$$